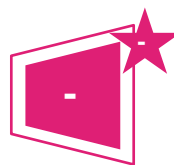




Asset Performance: Ene 08 Ventilation



Contributes to 40 credits
available for asset energy
rating

No Minimum Standard
rating

Aim

To minimise operational energy consumption and the associated carbon emissions by increasing the intrinsic energy efficiency of the building fabric and the installed buildings services

Question

1. What percentage of fans are fitted with variable speed drives (VSD)?
2. Are the pumps fitted with VSDs?
3. What is the specific fan power for air handling systems?
4. Has the ductwork been tested for leakage and appropriate remedial action taken?
5. In what year was the main ventilation system replaced (if known)?
6. What is the heat recovery rate of the ventilation system (if known)?

Credits	Question	For questions 1,3,5,6 enter value. For question 2 and 4 select either Yes or No	Units
-	1.	Enter value	%
-	2.	Yes/No	
-	3.	Enter value	W/l/s
-	4.	Yes/No	
-	5.	Enter year	
-	6.	Enter value	%

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.

Criteria	Evidence requirement
All	Extract of relevant O&M manuals or a copy of manufacturer information for the ventilation system.
All	Service/maintenance records for the ventilation system.
All	Visual inspection of the equipment date (photographic evidence if possible).
All	Copy of duct and air handling leakage test results.
5	Copy of documentation outlining when the ventilation system was installed or replaced, such as, a) Extract of O&M manuals or copy of manufacturer information for the ventilation system b) Service records c) Installation records d) Maintenance records.

Definitions

Air handling systems:

For the purposes of this issue, air handling systems are systems (usually centralised) which distribute air (usually for cooling) around the building.

Variable speed drive:

These are devices that can be attached to electrical motors to adjust the speed of a motor e.g., in a fan or a pump, when the service demand varies. These are also referred to as VFD (Variable frequency drive) and AFD (Adjustable frequency drive).

Additional information

Asset energy calculator

In order to obtain the maximum possible number of credits for the asset energy performance as much information as possible should be entered. Where no value is entered, the asset energy rating calculator will assume a pessimistic default value.



Asset Performance: Ene 09 Hot water



Contributes to 40 credits
available for asset energy
rating



No Minimum Standard

Aim

To recognise the contribution that hot water systems make to the overall energy performance of the asset.

Question

1. Is hot water generated on-site?
2. Is the main on-site hot water system local or centralised?
3. What is the main generation type for on-site hot water?
4. What energy source is used to generate on-site hot water?
5. Please enter the hot water generation efficiency or COP, (if known)
6. In what year was the main hot water generator/hot water system last replaced (if known)?
7. For hot water provided by district heating only;
 - a. Do you know what types of heat generation are used for the network heat supply?
 - b. What are the main heat generation sources?
 - c. What is the percentage distribution losses for the heat network (if known)?

Credits	Question	For questions 1 - 4, 7a - b select a single answer option. For question 5,6 and 7c enter value
-	1.	Yes / No
-	2.	Centralised / local
-	3.	Boiler Heat pump/ reversible chiller Direct electricity Other on-site system e.g. CHP/solar thermal Other (user entry)
-	4.	Question not answered Electric Burning Oil